Observable Dynamics: A Program

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1 Program overview

This document outlines a research program in which probabilistic, predictive, and dynamical structures are derived from observable experiments rather than assumed as primitive properties of an underlying state space.

The central thesis of the program is that the structures traditionally treated as foundational in probability and dynamical systems—states, laws, and dynamical evolution—may instead arise as consequences of observable compatibility and predictive organization.

In this perspective, observable experiments play the primary role, while probabilistic representations and dynamical structures appear as derived organizational principles.

Program scope. The goal of the observable dynamics program is not to replace classical probability or dynamical systems, but to provide an observational foundation from which probabilistic, predictive, and dynamical structure may arise as consequences of compatibility and prediction.

Classical formulations of probability and dynamical systems typically begin with a state space together with laws or evolution rules defined on that space. Observables are then treated as functions of the state. The present framework reverses this order of explanation. We begin with a class of admissible observational queries and their empirical regularities, and ask when probabilistic, predictive, and dynamical structures become unavoidable.

At a high level the program has the following architecture

$$\begin{aligned} &\text{observable experiments} \longrightarrow \text{probabilistic representation} \\ &\quad \longrightarrow \text{predictive state space} \longrightarrow \text{predictive operators.} \end{aligned}$$

More concretely, starting from a query system

$$(\mathcal{Q}, \preceq),$$

the foundational results show that compatible observable experiments admit a canonical probabilistic realization

$$(\Omega, \sigma(\mathcal{Q}), P).$$

Prediction then induces an equivalence relation on observable configurations yielding a minimal predictive query

$$Q_*,$$

whose states correspond to distinct predictive laws of future observables. Predictive evolution is then described by operators

$$(K_{Q_*}g)(q) = \mathbb{E}[g(F) \mid Q_* = q].$$

Thus the core mathematical spine of the program is

$$\{\ell_Q\} \xrightarrow{\text{P0}} (\Omega, P) \xrightarrow{\text{P1}} (\Omega, P) \xrightarrow{\text{P2}} Q_* \xrightarrow{\text{P3}} K_{Q_*}.$$

Here P0 derives σ-additivity from finitely additive marginals under SPUT; P1 assumes σ-additive marginals and establishes the canonical experiment; P2–P3 develop predictive structure and operator dynamics.

2 Program architecture

The observable dynamics program develops dynamical structure directly from observable experiments. The program proceeds through three conceptual stages.

Stage 0: Finitely additive foundations

finitely additive compatible family $\{\ell_Q\}$ $\longrightarrow$ SPUT $\longrightarrow$ σ-additive P on $(\Omega, \sigma(\mathcal{Q}))$

Paper: *Finitely Additive Observable Laws and the Prokhorov Extension for Query Systems*

Stage 1: Observational foundations

observable queries $\longrightarrow$ compatibility structure $\longrightarrow$ canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$

Paper: *Observational Foundations of Probability*

Stage 2: Predictive structure

(Ω, P) $\longrightarrow$ predictive kernels $\longrightarrow$ predictive equivalence $\longrightarrow$ minimal predictive query Q_*

Paper: *Predictive Experiments and Observable Operators*

Stage 3: Observable operator dynamics

Q_* $\longrightarrow$ predictive operator semigroup $\{K_t\}$ $\longrightarrow$ generator A, Kolmogorov equation

Paper: *Predictive Operator Theory for Observable Dynamical Systems*

Stage 4: Computational instantiation

delay query system $\longrightarrow$ predictive Markov order $m(\tau, P)$ $\longrightarrow$ local linear approximation K_t^N

Working note: *Observational Probability from Delay Queries*

Formal verification. Papers 1–3 are fully formalized in Lean 4 / Mathlib with zero `sorry` across all three source files (`QuerySystem.lean`, `PredictiveState.lean`, `PredictiveOperators.lean`). Paper 0 is nearly fully formalized (`ProkhorovExtension.lean`): the inverse limit theorem (`compactInverseLimit_r` and the main extension theorem (`prokhorov_extension`) are proved; one `sorry` remains in `prokhorov_extension_p` blocked on `PerfectMeasure` and Musiał's theorem being absent from Mathlib 4. See the Formalization Status section for details.

In summary, the program derives dynamical structure from observable experiments, progressing from observational compatibility through probabilistic representation, predictive state spaces, operator dynamics, and computational instantiation via delay embeddings.

3 Structure of the observable dynamics program

The observable dynamics program has a central theoretical spine together with complementary formal and constructive branches, as summarized in Figure 1.

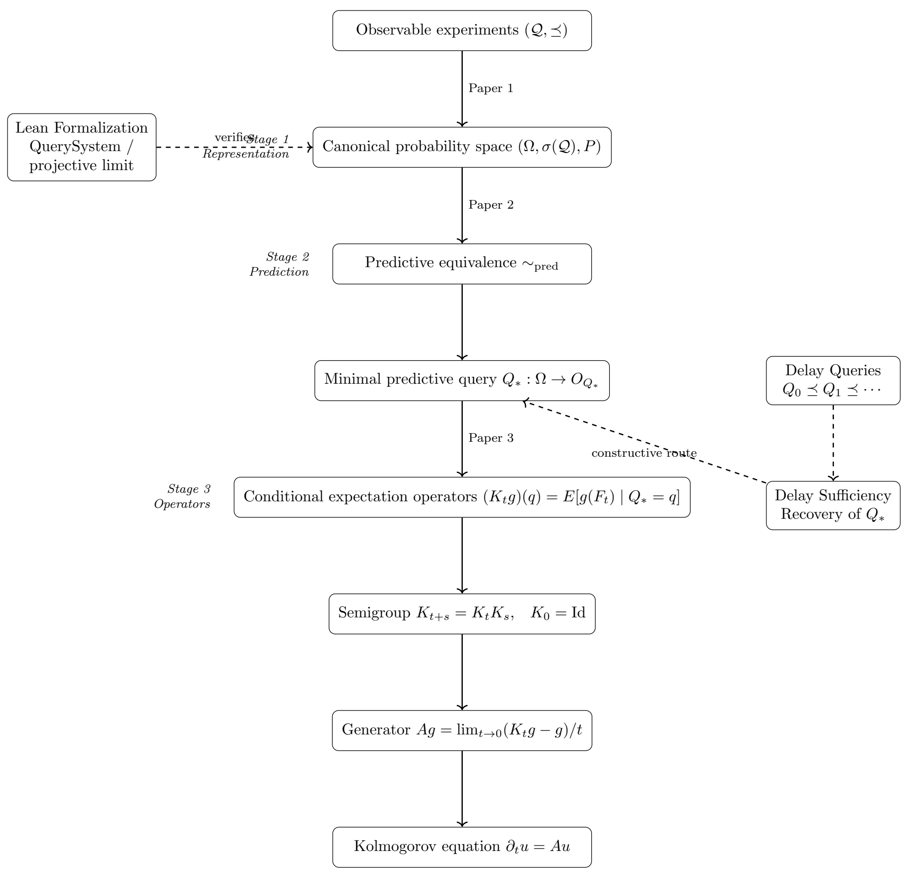

Figure 1: Derivational spine of the observable dynamics program. The main chain runs from observable queries through the canonical probability space, predictive equivalence, and the minimal predictive query Q_*, to the operator semigroup $\{K_t\}$, its generator A, and the Kolmogorov evolution equation. Lean formalization verifies the foundational layer (left). Delay-query constructions provide a constructive route to Q_* (right).

$$\textit{Observations} \to \textit{Probability} \to \textit{Prediction} \to \textit{State} \to \textit{Dynamics}$$

Observable experiments generate a probabilistic representation. Prediction then induces a minimal predictive state space, and predictive dynamics are represented by operators acting on functions of that state space.

Figure 1 displays the core architecture of the program. The first paper constructs probability from observable compatibility, the second constructs predictive state spaces from the canonical experiment, and the third develops the resulting operator dynamics.

In this sense the program is organized by a single structural progression:

$$\text{observations} \longrightarrow \text{probability} \longrightarrow \text{prediction} \longrightarrow \text{state} \longrightarrow \text{dynamics}.$$

4 Established theorems

The following theorems constitute the proved core of the program as of March 2026. All results in Stages 1–3 have Lean 4 / Mathlib proofs; Stage 4 results are proved in the working note.

4.1 Observational completion (Paper 1)

Theorem 1 (Observational Completion Theorem)**.** *Let $(\mathcal{Q}, \preceq)$ be a query system with compatible observable laws $\{\nu_Q\}_{Q\in\mathcal{Q}}$ satisfying the realizability and sequential upper-directedness conditions. Then there exists a probability space*

$$(\Omega, \sigma(\mathcal{Q}), P)$$

such that $Q_{\#}P = \nu_Q$ for every query Q. The space Ω is the projective limit of the outcome spaces; the σ-field $\sigma(\mathcal{Q})$ is generated by query-cylinder events; and P is the unique probability measure extending the compatible family.

Remark 1 (Determination)**.** *A companion uniqueness theorem (Observational Determination) shows that two probability measures on $(\Omega, \sigma(\mathcal{Q}))$ that agree on all query cylinders are identical, giving an observational analogue of the classical π-λ theorem.*

4.2 Predictive state and factorization (Paper 2)

Theorem 2 (Minimal Predictive Query)**.** *Let $(\Omega, \sigma(\mathcal{Q}), P)$ be the canonical experiment and $F : \Omega \to O_F$ a future observable with O_F standard Borel. Define*

$$Q_* := \phi_Q \circ Q : \Omega \to \mathcal{P}(O_F), \qquad \phi_Q(q) = P(F \in \cdot \mid Q = q).$$

Then Q_* is measurable, its states correspond exactly to distinct predictive laws of F, and the conditional law of F factors through Q_*:

$$P(F \in \cdot \mid Q_* = q_*) = q_*.$$

Moreover Q_* is universal: any measurable statistic R through which the predictive law factors satisfies $Q_* = \theta \circ R$ for a measurable θ, making Q_* the coarsest such statistic (Doob–Dynkin minimality).

Theorem 3 (Predictive Sufficiency)**.** *A query Q is predictively sufficient if and only if $F \perp Q' \mid Q$ for all queries Q'. The minimal predictive query Q_* is always predictively sufficient and is the unique minimal one up to predictive equivalence.*

4.3 Operator semigroup and duality (Paper 3)

Theorem 4 (Predictive Operator Semigroup)**.** *Assume the Markov kernels $\{P_t\}_{t\geq 0}$ on O_{Q_*} satisfy the Chapman–Kolmogorov equation. Then the operators*

$$(K_t g)(q_*) = \int g \, dP_t(q_*, \cdot)$$

form a Markov operator semigroup: $K_{t+s} = K_t K_s$, $K_0 = \mathrm{Id}$, each K_t positive and constant-preserving.

Theorem 5 (Koopman–Perron Duality)**.** *For finite measures μ on O_{Q_*} and bounded measurable g,*

$$\int (K_t g) \, d\mu = \int g \, d(P_t^* \mu),$$

where $P_t^* \mu(A) = \int P_t(q_*, A) \, \mu(dq_*)$. In the deterministic limit $K_t g = g \circ \phi_t$, recovering the classical Koopman operator, and P_t^* recovers the Perron–Frobenius transfer operator.

4.4 Delay queries and computational instantiation (Paper 4)

Theorem 6 (Upper-directedness of delay query system)**.** *The delay query system $\{Q_{d,\tau}\}$ with $Q_{d,\tau}(\omega) = (\omega_0, \omega_{-\tau}, \ldots, \omega_{-(d-1)\tau})$ is upper-directed: for any (d, τ) and (d', τ'), setting $\tau'' = \gcd(\tau, \tau')$ and $d'' = \max((d-1)\tau/\tau'', (d'-1)\tau'/\tau'') + 1$ gives a common refinement $Q_{d'',\tau''} \succeq Q_{d,\tau}, Q_{d',\tau'}$.*

Theorem 7 (Probabilistic Sufficiency Criterion)**.** *Define the predictive τ-Markov order*

$$m(\tau, P) = \min\{d \geq 1 : P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) = P(x_1 \in A \mid \mathcal{F}_\tau^-) \;\; P\text{-a.s. } \forall A\},$$

where $\mathcal{F}_\tau^- = \bigvee_{d=1}^\infty \sigma(\mathrm{eval}_{d,\tau})$. Then $Q_{d,\tau}$ is predictively sufficient if and only if $d \geq m(\tau, P)$. As a corollary, if the system has a d_M-dimensional compact manifold latent state, then $m(\tau, P) \leq 2d_M + 1$ (Takens specialization).

5 Structural principles

The observable-dynamics program is guided by several structural principles.

- **Completion principle.** Families of compatible observable experiments admit probabilistic representations.
- **Predictive sufficiency principle.** Prediction induces minimal observable state spaces capturing exactly the predictive information relevant to future observables.
- **Operator representation principle.** Predictive evolution admits operator-theoretic representations on predictive state spaces.
- **Forcing principle.** Compatibility constraints among observable regularities may force structural relations among observables.
- **Universality principle.** Distinct underlying mechanisms may exhibit identical observable regularities relative to a given query system.

6 Open problems and conjectures

The following distinguishes between the foundational stopping points — where the program's assumptions exceed its derivations — and longer-range structural conjectures about the program's broader reach.

Foundational stopping points

These are the honest gaps in the current theory: places where structure is assumed rather than derived. Resolving them would deepen the program's philosophical claim that probability arises from observable compatibility rather than being assumed.

Conjecture 1 (SP1: Measurable structure from discriminability)**.** *The σ-algebra $\mathcal{B}(O_Q)$ on each query outcome space is currently assumed. The program demands a derivation: starting from a primitive class $\mathcal{E}_Q$ of reportable distinctions (closed under finite Boolean operations), can σ-closure be forced by compatibility across refinements?*

Partial resolution (Paper -1, in development). *The question has been sharpened and partially resolved. First, a negative result: exhaustiveness — the first natural candidate for a finitary extension condition — is trivially satisfied by any normalized finitely-additive valuation, and carries no content. The genuine obstruction is* continuity at $\emptyset$ from above*: for decreasing sequences in $\mathcal{E}_Q$ whose intersection escapes $\mathcal{E}_Q$ into $\sigma(\mathcal{E}_Q) \setminus \mathcal{E}_Q$, the valuations must converge to zero. No finitary condition on any single query level can witness this; the regress is structural.*

Second, this is not a failure but a structural necessity: discriminability requires incompleteness. *An observer with $\mathcal{E}_Q = \sigma(\mathcal{E}_Q)$ has no horizon and admits no nontrivial refinement. The gap $\sigma(\mathcal{E}_Q) \setminus \mathcal{E}_Q$ is the constitutive outside that makes refinement possible. The correct reading of the program's chain is therefore:*

$$\textit{bounded discriminability} \;\rightarrow\; \textit{refinement} \;\rightarrow\; \textit{coherence} \;\rightarrow\; \textit{probability.}$$

The central open conjecture of Paper -1 is that sequential upper-directedness — already a hypothesis of the program — provides witnessing from above*: finer queries see what coarser ones cannot, and compatibility propagates that witnessing back to force continuity at $\emptyset$ at every finite level. If true, the σ-algebra is not derived by closing the gap but by the observer's commitment to acting as if a coherent world — one in which gaps close — exists, even though no observer in the system can occupy that completion.*

Conjecture 2 (SP2: σ-additivity from compatibility)**.** Closed in the compact setting (Paper 0, Corollary 6.3).

*Paper 0 (*Finitely Additive Observable Laws and the Prokhorov Extension for Query Systems*) resolves SP2 in two steps. First, under surjective projective uniform tightness (SPUT), any finitely compatible family of finitely additive valuations admits a unique σ-additive global extension (Theorem 6.1). Second, when each outcome space O_Q is compact Hausdorff and each refinement map is surjective, SPUT holds automatically with $K_Q := O_Q$ (Proposition 3.3). The architectural corollary (Corollary 6.3) therefore requires only finite compatibility and the topology of the query system:* the observational interface architecture alone forces countable additivity.

Closed in the standard Borel setting (Musia
l 1980). *When outcome spaces are standard Borel (measurably isomorphic to a Borel subset of a Polish space), SP2 is also resolved. Every Borel probability measure on a standard Borel space is perfect (Bogachev Vol. 2, §7.7.2), and Musiał's theorem (Fund. Math. 110, 1980) states that*

any projective system of perfect probability measures over a directed index set admits a unique σ-additive projective limit with $P(\Omega) = 1$, with no topological tightness assumption required. This covers all query systems whose outcome spaces are standard Borel: in particular, delay-embedding query systems with outcome spaces $\mathbb{R}^d$.

Open in the general non-compact, non-standard-Borel setting. *When outcome spaces are neither compact nor standard Borel (e.g. non-separable Banach spaces, general locally compact spaces), SPUT remains a genuine assumption. The purely structural or topology-free route to σ-additivity — deriving it from compatibility alone without tightness or perfectness hypotheses — is the remaining open form of SP2. Candidate approaches include Mallory–Sion-type arguments and compactification routes (one-point or Stone–Čech).*

Conjecture 3 (SP3: Realizability from compatibility)**.** *The projective limit Ω is assumed non-degenerate (realizability). The question is whether this follows from compatibility of the $\{\nu_Q\}$, or is genuinely independent. Candidate routes: almost-sure realizability derivable from compatibility alone; inner regularity (tightness) of the marginals implying a tight global limit.*

Conjecture 4 (SP4: Directedness from joint experiment formation)**.** *The three directedness conditions (lower, upper, sequential upper) are assumed axioms on $(\mathcal{Q}, \preceq)$. They should be derivable from an operational principle: if two experiments can always be performed jointly, upper-directedness follows. This reframes directedness as closure of the experimental repertoire under product formation rather than a poset axiom.*

Remark 2 (The compression mechanism)**.** *All four stopping points are connected by the compression argument at the heart of the extension theorem: a countable cover of a cylinder is compressed to a single query level, reducing a global statement to a local one. If compression can be made to force rather than merely use σ-additivity, measurability, and realizability, it becomes the engine of a fully observational derivation of probability.*

Categorical structure

Conjecture 5 (Adjunction between query and probability systems)**.** *There is an adjunction*

$$\mathrm{Ext} \dashv \mathrm{Forget} : \mathbf{ProbSys} \rightleftarrows \mathbf{QuerySys}$$

where Ext *sends a query system with compatible marginals to its canonical probability space, and* Forget *sends a probability space to its induced query system. Under the cofinality hypothesis of Paper 1, this adjunction becomes an equivalence of categories $\mathbf{QuerySys}^{\mathrm{cof}} \simeq \mathbf{ProbSys}$. This categorical statement has been sketched in exploratory notes (`categorical-foundations` branch) but not formally proved.*

Conjecture 6 (Functoriality of predictive state spaces)**.** *The assignment $(\mathcal{Q}, P) \mapsto Q_*$ is functorial: morphisms of query systems preserving predictive structure induce structure-preserving maps between minimal predictive state spaces and intertwine the predictive operators.*

Longer-range conjectures

Conjecture 7 (Universality of observable regularities)**.** *Distinct microscopic mechanisms may produce identical observable regularities relative to the same query system. Universality classes of query systems — systems with the same Q_* and operator spectrum — provide an observable classification of dynamical behavior that does not depend on the latent state.*

Conjecture 8 (Thermodynamic emergence). *Large-scale statistical laws (entropy, thermodynamic relations) arise from compatibility among observable experiments across observational scales. The observational framework provides a natural language for this: thermodynamic structure is the operator-spectral shadow of multi-scale compatibility.*

Conjecture 9 (Approximation bounds for K_t^N). *The local linear map truncation $K_t^N \to K_t$ in $L^2(\nu_{Q_*})$ has quantitative convergence rates depending on the geometry of O_{Q_*} and the mixing properties of P_t. Establishing explicit rates would connect the abstract operator theory to practical data-driven approximation.*

7 Motivation

The preceding sections state the program at a structural level. The purpose of the present note is to explain why this reversal of perspective is mathematically fruitful and why it leads naturally to a broader theory of observable dynamics.

Classical probability, dynamics, and statistical mechanics are typically formulated by starting from an assumed state space together with laws or evolution rules defined on that space. The observational framework reverses this order of explanation. One begins with a class of admissible queries and their observable regularities, and asks when probabilistic, predictive, and dynamical structure become unavoidable.

The broader question pursued here is what structures emerge once this observational starting point is adopted. These include recurrence and forcing, universality, thermodynamic organization, operator-theoretic structure, and exploratory applications to areas such as symbolic dynamics, statistical mechanics, and number theory.

Throughout, we assume a class of admissible boundary queries

$$\mathcal{Q},$$

and a corresponding family of observable regularities

$$\{\nu_Q\}_{Q\in\mathcal{Q}}.$$

No underlying path-space law or state-space dynamics is assumed.

8 Program architecture

8.1 Foundational layer: observational probability

The first paper of the program studies when a compatible family of observable experiments admits a canonical probabilistic realization. Schematically,

$$(\mathcal{Q}, \preceq) \;\longrightarrow\; (\Omega, \sigma(\mathcal{Q}), P).$$

Here $(\mathcal{Q}, \preceq)$ is a query system ordered by refinement, while $(\Omega, \sigma(\mathcal{Q}), P)$ is the canonical probabilistic representation whose query marginals recover the observed laws.

8.2 Predictive layer: minimal predictive state spaces

The second paper studies how predictive structure emerges once the canonical experiment has been constructed. Given a future observable, one obtains predictive kernels, predictive equivalence, a minimal predictive query Q_*, and predictive operators:

$$(\Omega, \sigma(\mathcal{Q}), P) \longrightarrow Q_* \longrightarrow K_{Q_*}.$$

Thus the first two papers establish the core pipeline

$$\text{observations} \longrightarrow \text{probability} \longrightarrow \text{predictive state} \longrightarrow \text{predictive operators.}$$

8.3 Broader program directions

Once this core architecture is in place, a number of additional research directions become natural:

- recurrence and forcing,
- universality and thermodynamic structure,
- symbolic and information-theoretic hierarchies,
- operator-theoretic formulations,
- data-driven and number-theoretic applications.

9 Current frontier

As of March 2026, the five-paper core spine is established and the foundational Paper -1 is under active development:

$$\mathcal{E}_Q \xrightarrow{\text{P}-1} \{\ell_Q\} \xrightarrow{\text{P0}} (\Omega, P) \xrightarrow{\text{P1}} Q_* \xrightarrow{\text{P2}} \{K_t\} \xrightarrow{\text{P3}} K_t^N \xrightarrow{\text{P4}} \text{(local linear)}.$$

Papers 0–4 are complete working papers with full proofs, abstracts, conclusions, related work, and bibliographies. The Lean formalization covers Papers 1–3 (zero `sorry`) and Paper 0 (one remaining `sorry` in `prokhorov_extension_polish`, Mathlib-blocked). Paper -1 (`discriminability_foundations`) is in early development: the negative result and the necessity of incompleteness are established; the central conjecture (witnessing from above via sequential upper-directedness) is open.

Immediate priorities.

1. **SP1: The witnessing conjecture (Paper -1).** The central open claim is that sequential upper-directedness provides witnessing from above for continuity at $\emptyset$, making σ-additivity a commitment rather than an assumption. The gap to close: showing that the colimit query's event algebra is rich enough to contain the relevant limit events. This is the most foundational open question in the program.

2. **SP2 in the general non-compact, non-standard-Borel setting.** SP2 is closed for compact (Paper 0, Corollary 6.3) and standard Borel (Musiał 1980) outcome spaces. The remaining open case is locally compact but non-standard-Borel. Note: a resolution of SP1 via the witnessing conjecture might simultaneously close this case, since the limit-query approach is topology-free.

3. **Lean formalization of Paper 0 (`prokhorov_extension_polish`).** One `sorry` remains, blocked on `PerfectMeasure` and Musiał's theorem being absent from Mathlib 4. No action until Mathlib upstream.

4. **Quantitative approximation bounds (Paper 4, Open Question 3).** Convergence $\|K_t^N - K_t\|_{L^2} \to 0$ is established but rates are unquantified. Bounds in terms of mixing rate, Fisher–Rao curvature, sample size N, and neighbourhood radius r would close the gap between theory and algorithm design.

10 Recurrence, forcing, and observable structure

10.1 Structural queries

Let $Q \in \mathcal{Q}$ be a boundary query with outcome space $\mathcal{O}_Q$. Typical examples include:

- finite-lag projections,
- coarse-grained symbolic encodings,
- delay embeddings,
- partition-based observations.

Each query produces an observable regularity

$$\nu_Q \in \mathcal{P}(\mathcal{O}_Q).$$

Definition 1 (Recurrence query)**.** *A query $Q \in \mathcal{Q}$ is called a* recurrence query *if its outcome space admits a natural notion of recurrence, such as:*

- *periodic symbolic patterns,*
- *return times,*
- *invariant or recurrent subsets,*
- *shift-invariant symbolic structures.*

Remark 3. *The definition is intentionally broad. The notion of recurrence is determined by the observational protocol rather than a privileged state-space structure.*

10.2 Observable recurrence structures

Let $\mathcal{S}_Q$ denote a collection of structural properties of ν_Q corresponding to recurrence phenomena. Examples include:

- existence of a periodic pattern of length n,
- positive entropy,
- mixing,
- existence of a subshift factor.

Definition 2 (Query-detectable structure)**.** *A recurrence structure $S \in \mathcal{S}_Q$ is said to be* query-detectable *if it depends only on the observable regularity ν_Q and not on any interior realization.*

Remark 4. *This ensures that recurrence is defined empirically, rather than as a property of a particular dynamical model.*

11 Boundary forcing and structural hierarchies

We now define a forcing relation among observable recurrence structures.

Definition 3 (Boundary forcing)**.** *Let S_1, S_2 be query-detectable recurrence structures (possibly arising from different queries). We say that*

$$S_1 \Rightarrow S_2$$

(S_1 boundary-forces S_2*) if the following holds:*

For every boundary horizon $\{\nu_Q\}$ that admits a completion compatible with the observed queries and satisfies structure S_1, every such completion also satisfies S_2.

Remark 5. *This relation depends only on observable regularities and their extendability, not on any assumed dynamical law.*

Remark 6. *Boundary forcing is therefore:*

- *query-relative,*
- *resolution-relative,*
- *independent of dimension,*
- *independent of state-space smoothness.*

Remark 7 (Relation to predictive state structure)**.** *Within the broader observational program, boundary forcing may be interpreted as a structural constraint on admissible predictive representations. In particular, forcing relations restrict the possible predictive state spaces and predictive operators that can arise from a given family of observable regularities.*

12 Classical and higher-dimensional recurrence applications

Definition 4 (Structural hierarchy)**.** *The partial order induced by boundary forcing defines a hierarchy of observable recurrence structures:*

$$S_1 \Rightarrow S_2.$$

This generalizes classical forcing relations in dynamical systems.

Proposition 1. *Boundary forcing is transitive and reflexive.*

Proof. Reflexivity is immediate.

If $S_1 \Rightarrow S_2$ and $S_2 \Rightarrow S_3$, then any compatible completion exhibiting S_1 exhibits S_2, and therefore S_3. □

13 Classical Sharkovskii ordering as a special case

We now show that the classical Sharkovskii theorem fits into this framework.

Let $Y = [0, 1]$ and consider discrete maps

$$f : Y \to Y.$$

Choose a partition-based symbolic query:

$$Q : \text{orbit} \mapsto \text{symbolic itinerary}.$$

Define structures:

- S_n: existence of a symbolic periodic pattern of period n.

Theorem 8 (Sharkovskii as boundary forcing)**.** *In one-dimensional interval dynamics, the classical Sharkovskii ordering corresponds to a boundary forcing hierarchy among symbolic periodic structures.*

Remark 8. *This interpretation does not depend on differentiability or smoothness, but only on observable symbolic recurrence.*

14 Higher-dimensional generalization

In higher dimensions, periodicity alone is insufficient to capture forcing. Instead, the relevant structures include:

- subshifts of finite type,
- topological entropy,
- invariant symbolic factors,
- braid or homological recurrence.

Definition 5 (Symbolic dominance)**.** *We say that structure S_1 symbolically dominates S_2 if any observable realization of S_1 induces S_2 under the same query class.*

This leads to a generalized Sharkovskii-type hierarchy based on symbolic and information-theoretic structure rather than period alone.

15 Universality, equilibrium, and thermodynamic structure

The next layer of the program concerns the emergence of large-scale statistical organization from compatibility and completion constraints. The main idea is that familiar structures such as equilibrium, universality, and thermodynamic behavior may be interpreted as consequences of observable consistency rather than as primitive assumptions.

15.1 Program directions

This part of the program includes:

- query-relative entropy hierarchies,
- information-geometric forcing,
- statistical-mechanical interpretations of recurrence,
- empirical detection of structural transitions.

In particular, the boundary completion framework suggests that structural forcing is fundamentally a question of extendability across resolutions.

15.2 Exploratory application: the Riemann zeta function

This note outlines a possible research program for applying the boundary-query framework to number-theoretic structures, with particular focus on the Riemann zeta function. The intent is exploratory: no structural assumptions are imposed a priori, and all dynamical or operator structure is treated as derived.

15.3 Motivation

The Riemann zeta function

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

admits a formal analogy with dynamical zeta functions arising in chaotic systems, where periodic orbits replace prime numbers. This suggests the possibility of an underlying dynamical or statistical structure. However, most existing approaches assume such a structure in advance.

Within the observational program, one may instead ask whether dynamical or operator-theoretic structure could emerge from observable regularities of prime statistics across scales.

15.4 Observable regularities of prime sequences

Let $\mathbb{P}$ denote the ordered sequence of prime numbers. Rather than starting with an analytic function, we consider a class of observable queries $\mathcal{Q}$ acting on this sequence.

Examples include:

- Gap statistics: distributions of $p_{n+1} - p_n$.
- Local correlation functions.
- Windowed prime-count statistics.
- Spectral statistics of truncated Dirichlet series.
- Symbolic encodings of primes.

For each query $Q \in \mathcal{Q}$, we define a set of admissible empirical regularities $\mathcal{R}_Q$. A boundary horizon is then a family

$$\{\nu_Q\}_{Q \in \mathcal{Q}}, \qquad \nu_Q \in \mathcal{R}_Q,$$

interpreted as the observable regularities of the prime sequence at various scales.

15.5 Boundary completion across scales

We ask whether these observable regularities admit approximate extension across scales. In particular, for queries defined at resolutions n and m, we study whether joint realizability holds:

$$\nu_{Q_{n+m}} \approx \mathrm{Ext}(\nu_{Q_n}, \nu_{Q_m}).$$

Failure of such completion identifies regimes where multi-scale prime statistics cannot be consistently unified.

Approximate completion provides a quantitative measure of coherence across scales and may be related to conjectures such as:

- Montgomery pair correlation.

- Random matrix universality.
- Distribution of zeros.

15.6 Statistical forcing and universality

A central question is whether certain low-resolution regularities force higher-order structure. For example:

- Do local gap statistics constrain long-range correlations?
- Is random-matrix behavior forced by compatibility across scales?

This suggests the possibility of forcing hierarchies among observable prime statistics analogous to recurrence hierarchies in dynamical systems.

15.7 Emergence of transfer operators

If boundary regularities admit suitable disintegrations, it may be possible to obtain kernel-based predictive representations:

$$T_{\Delta t} : \mathcal{X} \rightsquigarrow \mathcal{X},$$

leading potentially to transfer-operator formulations.

The key point is that such operators are not assumed but derived. This may provide a pathway toward:

- Spectral interpretations of ζ.
- Connections to the Hilbert–Pólya conjecture.
- Thermodynamic formalism for primes.

15.8 Non-equilibrium statistical mechanics viewpoint

From the present perspective, the zeta function may be viewed as encoding multi-scale regularities of the prime sequence rather than solely as an analytic object.

This connects to:

- Large deviation theory.
- Entropy and information flow.
- Renormalization and scaling.

One speculative possibility is that classical conjectures such as the Riemann hypothesis could correspond to structural stability conditions for multi-scale compatibility of observable regularities.

15.9 Initial concrete program

A feasible first project is the study of approximate boundary completion for prime gap statistics:

1. Define a hierarchy of finite-resolution queries.
2. Measure extension consistency across scales.
3. Compare with predictions from random matrix theory.

Such investigations may clarify which statistical regularities are fundamental and which emerge as consequences of multi-scale compatibility.

15.10 Outlook

This perspective aims to unify number theory, statistical mechanics, and dynamical systems within a common empirical framework. The goal is not to impose dynamics on prime structure but to determine when and how such dynamics become unavoidable.

Further directions include:

- Higher-dimensional symbolic models of primes.
- Connections to quantum chaos.
- Operator-theoretic constructions guided by boundary completion.

16 Universality and thermodynamic structure

We now outline how large-scale statistical and dynamical structure may arise as consequences of compatibility and extension constraints on boundary regularities. The viewpoint adopted here is that familiar notions such as equilibrium, universality, and thermodynamic behavior need not be assumed a priori, but may instead emerge as consequences of boundary consistency.

16.1 Boundary forcing

Remark 9 (Structural forcing viewpoint)**.** *The concept of boundary forcing introduced earlier may be interpreted more generally as a structural principle: compatibility constraints among observable regularities restrict the class of admissible realizations. In this way, dynamical or thermodynamic structure may appear as a necessary consequence of empirical consistency rather than as a modeling assumption.*

Typical examples of structural properties include:

- existence of kernel representations,
- approximate semigroup structure,
- existence of a generator,
- mixing or ergodic behavior,
- scaling or universality properties.

Thus, dynamics appears not as a primitive assumption but as a necessary consequence of compatibility among observable regularities.

Remark 10 (Forcing versus modeling). *Boundary forcing is independent of modeling choices. It identifies structural features that are unavoidable once empirical constraints are imposed.*

16.2 Universality as stability of boundary structure

Universality refers to the empirical phenomenon that distinct microscopic models produce indistinguishable large-scale statistical behavior. Within the present framework, this is interpreted as stability of boundary regularities under admissible realizations.

Definition 6 (Boundary universality). *A class of realizations is said to exhibit* boundary universality *if they induce the same boundary regularities for a specified class of queries.*

This viewpoint shifts emphasis from interior structure to observable regularities. Universality emerges when extension and compatibility constraints restrict the admissible boundary families to a small set.

Remark 11. *Universality therefore reflects non-uniqueness of interior realizations consistent with the same boundary horizon.*

16.3 Equilibrium as exact boundary completion

Exact boundary completion across resolutions provides a natural notion of equilibrium.

Definition 7 (Equilibrium boundary horizon). *A boundary horizon is said to be in* equilibrium *if the associated boundary family is exactly complete across all resolutions.*

In this case, finite-resolution regularities are mutually consistent and admit coherent extension at all scales.

Conversely, deviations from equilibrium correspond to structured failure of boundary completion.

Remark 12 (Non-equilibrium regimes). *Non-equilibrium behavior may be interpreted as persistent incompatibility of boundary regularities across resolutions. Such incompatibility can arise even when each individual scale appears well-behaved.*

16.4 Approximate completion and large deviations

Approximate boundary completion naturally leads to a large deviation interpretation of empirical incompatibility.

Let ε denote the minimal divergence required to reconcile boundary regularities across scales. Then ε may be interpreted as a cost of enforcing consistency.

Definition 8 (Boundary rate function). *A rate function associated with boundary incompatibility is defined by*

$$I := \inf_{\widetilde{\Gamma}\in \mathrm{Ext}(\Gamma_1,\Gamma_2)} D_f(\Gamma_{12}, \widetilde{\Gamma}),$$

where Γ_{12} denotes an observed long-lag regularity.

This quantity measures the minimal empirical cost of reconciling observed regularities with a jointly realizable structure.

Remark 13. *This interpretation connects boundary completion with large deviation theory. Compatibility corresponds to zero cost, while deviations correspond to rare or constrained behavior.*

16.5 Thermodynamic formalism as an emergent structure

Thermodynamic formalism arises when boundary forcing selects a preferred class of compatible realizations.

In particular:

- entropy quantifies multiplicity of compatible realizations,
- free energy reflects trade-offs between compatibility and constraint,
- equilibrium measures maximize compatibility subject to constraints.

Thus, thermodynamic structure is interpreted as an optimization principle arising from empirical consistency rather than as an intrinsic property of a predefined dynamical system.

Remark 14 (Maximum entropy and least action)**.** *Principles such as maximum entropy and least action may be viewed as selection criteria among admissible realizations that maintain compatibility across scales.*

16.6 Toward universality classes of boundary horizons

A long-term goal of this framework is to classify boundary horizons according to their forcing and universality properties.

Such a classification would parallel:

- phase transitions in statistical mechanics,
- universality classes in critical phenomena,
- invariant structures in dynamical systems.

In this sense, the boundary horizon provides a common language for relating empirical regularities, statistical structure, and dynamical representations.

17 Technical scaffolding for the program

The preceding sections describe the conceptual program and several application families. The present section records a possible categorical organization of the framework. Its role is not to drive the main narrative of the note, but to clarify how the observational completion construction sits between classical probabilistic models and more general query-based observational specifications.

17.1 Formal categorical scaffolding for the query/compatibility framework

Definition 9 (Category **Meas**)**.** *Objects are measurable spaces (X, Σ_X). Morphisms are measurable maps $f : (X, \Sigma_X) \to (Y, \Sigma_Y)$.*

Definition 10 (Category **Prob**)**.** *Objects are pairs (X, μ) where X is standard Borel and $\mu \in \mathcal{P}(X)$. A morphism $(X, \mu) \to (Y, \nu)$ is a measurable map $f : X \to Y$ such that $f_{\#}\mu = \nu$.*

Definition 11 (Category **Stoch** (optional))**.** *Objects are standard Borel spaces. A morphism $X \rightsquigarrow Y$ is a Markov kernel $K : X \times \mathcal{B}(Y) \to [0, 1]$. Composition is kernel composition. (Equivalently,* **Stoch** *is the Kleisli category of the Giry monad.)*

Definition 12 (Directed preorders and projective systems in **Meas**)**.** *Let $(\mathcal{Q}, \preceq)$ be a directed preorder (for any finite $Q_1, \dots, Q_n$ there exists $Q_\star$ with $Q_i \preceq Q_\star$). A* projective system of measurable spaces *indexed by $\mathcal{Q}$ is:*

- *objects $(O_Q, \mathcal{B}(O_Q))$ for each $Q \in \mathcal{Q}$,*
- *measurable bonding maps $\pi^{Q_2}_{Q_1} : O_{Q_2} \to O_{Q_1}$ for $Q_1 \preceq Q_2$,*

satisfying identity and cocycle:

$$\pi^{Q}_{Q} = \mathrm{id}_{O_Q}, \qquad \pi^{Q_3}_{Q_1} = \pi^{Q_2}_{Q_1} \circ \pi^{Q_3}_{Q_2} \quad (Q_1 \preceq Q_2 \preceq Q_3).$$

Definition 13 (Compatible family of laws)**.** *Given a projective system $\{O_Q, \pi\}$, a family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ with $\nu_Q \in \mathcal{P}(O_Q)$ is* projectively compatible *if*

$$(\pi^{Q_2}_{Q_1})_\# \nu_{Q_2} = \nu_{Q_1} \qquad \forall\, Q_1 \preceq Q_2.$$

Remark 15 (The fundamental commutative square)**.** *For $Q_1 \preceq Q_2$, compatibility is the statement that the following diagram commutes in* **Prob***:*

$$\begin{tikzcd}
(O_{Q_2}, \nu_{Q_2}) \arrow[r, "\pi^{Q_2}_{Q_1}"] \arrow[dr, "\mathrm{id}"'] & (O_{Q_1}, \nu_{Q_1}) \arrow[d, "\mathrm{id}"] \\
& (O_{Q_1}, \nu_{Q_1})
\end{tikzcd} \qquad \textit{equivalently} \qquad (\pi^{Q_2}_{Q_1})_\# \nu_{Q_2} = \nu_{Q_1}.$$

Definition 14 (Category **ObsSpec** of observational specifications)**.** *An object is a triple*

$$\mathsf{S} = \big(\mathcal{Q}, \{O_Q\}_{Q\in\mathcal{Q}}, \{\pi^{Q_2}_{Q_1}\}_{Q_1\preceq Q_2}; \{\nu_Q\}_{Q\in\mathcal{Q}}\big)$$

where $(\mathcal{Q}, \preceq)$ is directed, $\{O_Q, \pi\}$ is a projective system of standard Borel spaces, and $\{\nu_Q\}$ is projectively compatible.

A morphism $\mathsf{S} \to \mathsf{S}'$ consists of:

- *a monotone map $\alpha : (\mathcal{Q}, \preceq) \to (\mathcal{Q}', \preceq')$,*
- *measurable maps $\theta_Q : O_Q \to O'_{\alpha(Q)}$ for each Q,*

such that the squares commute for all $Q_1 \preceq Q_2$:

$$\begin{tikzcd}
O_{Q_2} \arrow[r, "\pi^{Q_2}_{Q_1}"] \arrow[d, "\theta_{Q_2}"'] & O_{Q_1} \arrow[d, "\theta_{Q_1}"] \\
O'_{\alpha(Q_2)} \arrow[r, "\pi'^{\alpha(Q_2)}_{\alpha(Q_1)}"'] & O'_{\alpha(Q_1)}
\end{tikzcd}$$

and the laws are preserved:

$$(\theta_Q)_\# \nu_Q = \nu'_{\alpha(Q)} \qquad \forall\, Q \in \mathcal{Q}.$$

Remark 16. **ObsSpec** *captures "laws of observations + refinement" without positing an underlying sample space at all.*

Definition 15 (Projective limit (thread) space)**.** *Given* $\mathsf{S} \in \mathbf{ObsSpec}$*, define the thread space*

$$O_\infty := \left\{ (x_Q)_{Q\in\mathcal{Q}} \in \prod_{Q\in\mathcal{Q}} O_Q : x_{Q_1} = \pi^{Q_2}_{Q_1}(x_{Q_2}) \ \forall\, Q_1 \preceq Q_2 \right\}.$$

Let $\mathrm{pr}_Q : O_\infty \to O_Q$ *denote coordinate projections.*

Proposition 2 (Kolmogorov–Carathéodory on projective limits (abstract form))**.** *If all* O_Q *are standard Borel and* $\mathcal{Q}$ *is countably generated (or there exists a countable cofinal sub-preorder), then any compatible family* $\{\nu_Q\}$ *induces a unique probability measure* $\nu_\infty \in \mathcal{P}(O_\infty)$ *such that*

$$(\mathrm{pr}_Q)_\# \nu_\infty = \nu_Q \qquad \forall\, Q \in \mathcal{Q}.$$

Definition 16 (Completion functor $\mathcal{C} : \mathbf{ObsSpec} \to \mathbf{Prob}$)**.** *On objects:* $\mathcal{C}(\mathsf{S}) := (O_\infty, \nu_\infty)$*, where* ν_∞ *is the unique thread-space measure with the prescribed marginals.*

On morphisms: given $(\alpha, \{\theta_Q\}) : \mathsf{S} \to \mathsf{S}'$*, define* $\Theta : O_\infty \to O'_\infty$ *by* $(x_Q)_Q \mapsto (\theta_{\bar{Q}}(x_{\bar{Q}}))_{\bar{Q}\in\mathcal{Q}'}$ *using a cofinal choice / induced map (details depend on whether* α *is cofinal). Under the standard hypotheses,* Θ *is measurable and satisfies* $\Theta_\# \nu_\infty = \nu'_\infty$*; set* $\mathcal{C}(\alpha, \theta) := \Theta$*.*

Remark 17 (What is fully canonical)**.** *The object-level construction* $\mathsf{S} \mapsto (O_\infty, \nu_\infty)$ *is canonical. Functoriality on morphisms is cleanest when one restricts to cofinal monotone maps* α *(or works in an equivalent "pro-object" formulation). This is a technicality of index management, not of the completion principle itself.*

Definition 17 (Representations of an observational specification)**.** *Fix a standard Borel space* Ω*. A* representation *of* S *on* Ω *is a family of measurable maps* $Q : \Omega \to O_Q$ *such that for all* $Q_1 \preceq Q_2$*,*

$$Q_1 = \pi^{Q_2}_{Q_1} \circ Q_2.$$

Equivalently, there is a unique evaluation map

$$\Phi : \Omega \to O_\infty, \qquad \Phi(\omega) := (Q(\omega))_{Q\in\mathcal{Q}}$$

with $\mathrm{pr}_Q \circ \Phi = Q$ *for all* Q*.*

Definition 18 (Query-generated σ-algebra on Ω)**.** *Given a representation* $\{Q : \Omega \to O_Q\}$*, define*

$$\sigma(\mathcal{Q}) := \sigma\big(\{Q^{-1}(B) : Q \in \mathcal{Q},\ B \in \mathcal{B}(O_Q)\}\big).$$

Proposition 3 (Pullback model (your "measure on Ω"))**.** *Let* $\mathsf{S} \in \mathbf{ObsSpec}$ *and let* $\{Q : \Omega \to O_Q\}$ *be a representation. Let* $(O_\infty, \nu_\infty) = \mathcal{C}(\mathsf{S})$ *with evaluation map* $\Phi : \Omega \to O_\infty$*. Then*

$$\mathbb{P} := \Phi^{-1}_\# \nu_\infty$$

is a probability measure on $(\Omega, \sigma(\mathcal{Q}))$ *satisfying*

$$Q_\# \mathbb{P} = \nu_Q \qquad \forall\, Q \in \mathcal{Q}.$$

If Φ *is a measurable isomorphism from* $(\Omega, \sigma(\mathcal{Q}))$ *onto* $(O_\infty, \mathcal{B}(O_\infty))$ *(equivalently:* $\sigma(\mathcal{Q})$ *separates points up to* ν_∞*-null sets), then* $\mathbb{P}$ *is the unique such measure on* $(\Omega, \sigma(\mathcal{Q}))$*.*

Remark 18 (Relation to the observational program)**.** *The categorical construction above provides a precise formulation of the observational completion principle. Observable specifications determine a canonical probabilistic representation, while concrete state-space models appear as representations of that canonical object.*

Definition 19 (Coordinate experiment subcategory **CoordSpec** $\subset$ **ObsSpec**)**.** *Let $\mathcal{Q}$ be the family of finite coordinate projections $\pi_F : Y^{\mathbb{Z}} \to Y^F$ indexed by finite $F \subset \mathbb{Z}$, ordered by inclusion. Set $O_{\pi_F} := Y^F$ and $\pi^{\pi_G}_{\pi_F} : Y^G \to Y^F$ as coordinate restriction when $F \subset G$. Objects of* **CoordSpec** *are compatible families $\{\mu_F\}_{F \subset \mathbb{Z},\ |F|<\infty}$.*

Remark 19. *Then $\mathcal{C}$ restricted to* **CoordSpec** *recovers the classical Kolmogorov extension: the thread space O_∞ is canonically $Y^{\mathbb{Z}}$ and ν_∞ is the process law on the product σ-algebra.*

Remark 20 (Category-level diagram)**.** *The relationship can be summarized as:*

$$\begin{tikzcd}
\mathbf{CoordSpec} \arrow[r, hook] \arrow[d, "\mathcal{C}|_{\mathbf{CoordSpec}}"'] & \mathbf{ObsSpec} \arrow[d, "\mathcal{C}"] \\
\mathbf{Prob} \arrow[r, equal] & \mathbf{Prob}
\end{tikzcd}$$

where $\mathcal{C}$ is the canonical completion to the projective-limit (thread) probability space. Representations on a chosen carrier Ω are additional structure (a map $\Phi : \Omega \to O_\infty$) and are not part of $\mathcal{C}$ itself.

Definition 20 (Observable-extraction assignment (not fully functorial without choices))**.** *Given $(X, \mu) \in$* **Prob** *one can form an observational specification by selecting a directed family of measurable maps $Q : X \to O_Q$ and taking $\nu_Q := Q_{\#}\mu$. This is a* choice *of observational interface, not canonical unless a class of queries is fixed.*

Remark 21. *This is why one should not claim a naive adjunction* **ObsSpec** $\rightleftarrows$ **Prob** *without specifying a query-selection policy. What* is *canonical is:*

$$\mathbf{ObsSpec} \xrightarrow{\mathcal{C}} \mathbf{Prob}$$

and then "models on Ω" are representations (pullbacks) of $\mathcal{C}(\mathsf{S})$.

18 Program questions

Several natural questions arise from the observable-dynamics framework.

- Under what general conditions do predictive state spaces exist?
- How large must a query system be for predictive sufficiency to emerge?
- What structural constraints do observable compatibility conditions impose on predictive operators?
- To what extent do classical dynamical invariants arise as properties of predictive operator spectra?
- Can universality classes of observational systems be classified in terms of compatibility structures?

These questions suggest that observable experiments may provide a natural organizing language for probabilistic and dynamical theory.

19 Program outlook

The overall research program may now be summarized schematically as

$$\begin{aligned}&\text{observable experiments} \longrightarrow \text{probabilistic representation} \longrightarrow \text{predictive state spaces}\\ &\quad\longrightarrow \text{predictive operators} \longrightarrow \text{structural applications.}\end{aligned}$$

The first two stages are developed in the foundational paper *Observational Foundations of Probability* and the sequel *Predictive Experiments and Observable Operators.* The remaining stages organize the broader research agenda: recurrence and forcing, universality and thermodynamics, operator-theoretic emergence, and exploratory applications such as symbolic dynamics, statistical mechanics, and number theory.

In this way the program aims to provide a unified observational route from empirical regularities to probabilistic, predictive, and dynamical structure. Its central thesis is that states and dynamics need not be assumed in advance: they may instead emerge from the compatibility and predictive organization of observable experiments.

20 Formalization status

Updated March 2026. The Lean source files are located at `formalization/QuerySystem/QuerySystem/` in the repository.

The following summarizes the current state of the Lean 4/Mathlib formalization of Papers 1–3 (`QuerySystem.lean`, `PredictiveState.lean`, `PredictiveOperators.lean`). Paper 4 (delay queries, computational instantiation) is not formalized; it exists as a working note with pen-and-paper proofs.

Fully formalized

The following theorems have complete Lean proofs with zero `sorry`:

- *Paper 1*: observational determination theorem (measure and probability versions); cylinder family π-system property; premeasure well-definedness and finite additivity; structural infrastructure for the observational extension theorem.

- *Paper 2*: predictive kernel construction (via Rokhlin disintegration); measurability of the predictive law map ϕ_Q; predictive compatibility (tower property, a.e. form); minimal predictive query Q_* and its measurability; predictive factorization theorem; predictive sufficiency; canonical operator properties (positivity, constant preservation, linearity, contraction, measurability).

- *Paper 3*: semigroup identity, semigroup property $K_{t+s} = K_t K_s$ (discrete time); Markov constant preservation and positivity; Koopman–Perron duality; deterministic specialization to classical Koopman operators; deterministic semigroup flow law.

Assumptions the formalization requires

Four hypotheses appear in Lean but are not always explicit in the papers:

- The future observable outcome space O_F must be a *standard Borel space* and *nonempty* for the Rokhlin disintegration (`condKernel`) to exist.

- Observable functions g must be *bounded* measurable for the factorization, sufficiency, semigroup, and duality theorems.
- The deterministic semigroup law requires the state space to *separate points* (for injectivity of the Dirac map).
- Koopman–Perron duality requires a *finite* (not merely σ-finite) input distribution μ.

Remaining deferred obligation

One intentional gap remains: the σ*-subadditivity* of the premeasure on finite cylinder events (`cylGen_addContent_isSigmaSubadditive` in `QuerySystem.lean`). This is the analytic core of the Carathéodory extension step: to pass from finite additivity of the observational content to a countably additive probability measure on $(\Omega, \sigma(\mathcal{Q}))$, one must verify that the premeasure satisfies $\mu_0(\bigcup_n C_n) \leq \sum_n \mu_0(C_n)$ for cylinder sequences decreasing to $\emptyset$. The topological hypotheses of the observational extension theorem (Polish spaces, Radon measures, countable cofinality) are precisely the conditions that make this argument available, but the Lean proof has not yet been completed. All algebraic and measurability infrastructure surrounding this step is in place.